

CBSE Class 12
Computer Science
Previous Year Question Paper 2015

Series: SS0

Set- 4

Code no. 91

- Please check that this question paper contains 15 printed pages.
- Code number given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 7 questions.
- **Please write down the Serial Number of the question before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

COMPUTER SCIENCE

Time allowed: 3 hours

Maximum Marks: 70

Instructions:

- All questions are compulsory.
- Programming Language: C++

SECTION - A

Only for candidates, who opted for C++

1. (a) Find the correct identifiers out of the following, which can be used for naming variables, Constants or Functions in a C++ program. For, while, INT, NeW, delete, 1stName, Add+Subtract, name1. 2 Marks

Ans: For, INT, NeW, name1 could be used as the identifiers. All others are predefined keywords used in a program.

(b) Observe the following program very carefully and write the names of those header file(s), which are essentially needed to compile and execute the following program successfully: 1 Mark

```
typedef char STRING[80];

void main()
{
    STRING Txt[] = "We love Peace";
    int Count=0;
    while (Txt[Count]!='\0')
        if (isalpha(Txt[Count]))
            Txt[Count++]='@';
        else
            Txt[Count++]='#';
    puts(Txt);
}
```

Ans: ctype and stdio are the required header files in the above code fragment.

(c) Observe the following C++ code very carefully and rewrite it after removing any/all syntactical errors with each correction underlined. 2 Marks

Note: Assume all required header files are already being included in the program.

#Define float MaxSpeed=60.5;

void main()

```
{
    int MySpeed
    char Alert='N';
    cin>>MySpeed;
    if MySpeed>MaxSpeed
        Alert='Y';
    cout<<Alert<<endl;
}
```

Ans:

#define float MaxSpeed=60.5; _____ // 3 errors

void main()

```
{
    int MySpeed; _____ // 1 error
    char Alert='N';
    cin>>MySpeed;
    if (MySpeed>MaxSpeed) //1 error
        Alert='Y';
    cout<<Alert<<endl; //1 error
}
```

(d) Write the output of the following C++ program code:

2 Marks

Note : Assume all required header files are already being included in the program.

```
void Location(int &X,int Y=4)
```

```
{
```

```
    Y+=2;
```

```
    X+=Y;
```

```
}
```

```
void main()
```

```
{
```

```
    int PX=10,PY=2;
```

```
    Location(PY);
```

```
    cout<<PX<<","<<PY<<endl;
```

```
    Location(PX,PY);
```

```
    cout<<PX<<","<<PY<<endl;
```

```
}
```

Ans: The output of this C++ program is:

10,8

20,8

(e) Write the output of the following C++ program code:

2 Marks

Note: Assume all required header files are already being included in the program.

```
class Eval
```

```
{  
  
    char Level;  
  
    int Point;  
  
    public:  
    Eval()  
    {  
        Level='E';  
        Point=0;  
    }  
  
    void Sink(int L)  
    {  
        Level=L;  
    }  
  
    void Float(int L)  
    {  
        Level+=L;  
        Point++;  
    }  
  
    void Show()  
    {  
        cout<<Level<<"#"<<Point<<endl;  
    }  
}
```

```
};  
  
void main()  
{  
    Eval E;  
    E.Sink(3);  
    E.Show();  
    E.Float(7);  
    E.Show();  
    E.Sink(2);  
    E.Show();  
}
```

Ans: Its output is:

B#0

I#1

G#1

(f) Study the following program and select the possible output(s) from the options (i) to (iv) following it. Also, write the maximum and the minimum values that can be assigned to the variable VAL. 2 Marks

Note:

Assume all required header files are already being included in the program.

random(n) function generates an integer between 0 and n-1.

```
void main()  
{
```

```

randomize();

int VAL;

VAL=random(3)+2;

char GUESS[]="ABCDEFGHIJK";

for (int I=1;I<=VAL; I++)
{
    for(int J=VAL; J<=7;J++)
        cout<<GUESS[J];
    cout<<endl;
}
}

```

(i)	(ii)	(iii)	(iv)
BCDEFGH	CDEFGH	EFGH	FGHI
BCDEFGH	CDEFGH	EFGH	FGHI
		EFGH	FGHI
		EFGH	FGHI

Ans: Of the given outputs (ii) and (iii) are correct.

Min and Max values of VAL are 2 & 4

2. (a) What is a copy constructor ? Give a suitable example in C++ to illustrate with its definition within a class and a declaration of an object with the help of it. 2 Marks

Ans: A copy constructor is the member function of the class which is also known as an overloaded constructor. In this the object of the same class is passed as a reference parameter.

For eg:

Class Copy

```
{int i;
```

```
    public:
```

```
    Copy ()
```

```
    {i=0;
```

```
    }
```

```
    Copy (Copy &c)    //Copy Constructor
```

```
    { i = c*i;
```

```
    }
```

```
};
```

```
void main ()
```

```
{Copy c1;
```

```
    Copy c2(c1);    //Copy Constructor is invoked
```

```
}
```

(b) Observe the following C++ code and answer the questions (i) and (ii) :

```
class Passenger
```

```
{long PNR;
```

```

char Name[20];

public:

    Passenger()                                //Function 1
    {
        cout<<"Ready"<<endl;
    }

    void Book(long P,char N[])                //Function 2
    {
        PNR = P; strcpy(Name, N);
    }

    void Print()                              //Function 3
    {
        cout<<PNR << Name <<endl;
    }

    ~Passenger()                             //Function 4
    {
        cout<<"Booking cancelled!"<<endl;
    }

};

```

(i) Fill in the blank statements in Line 1 and Line 2 to execute Function 2 and Function 3 respectively in the following code: **1 Mark**

```

void main()
{
    Passenger P;
    _____ //Line 1
    _____ //Line 2
} //Ends here

```

Ans: P.Book(1111111,"Ram"); {In Line 1}

P.Print(); {In Line 2}

(ii) Which function will be executed at } // Ends here ? What is this function referred to as ?

Ans: Function 4 will be executed as it is a destructor function.

(c) Write the definition of a class Photo in C++ with following description:

4 Private Members

- Pno //Data member for Photo Number (an integer)
- Category //Data member for Photo Category (a string)
- Exhibit //Data member for Exhibition Gallery (a string)
- FixExhibit //A member function to assign

//Exhibition Gallery as per Category

//as shown in the following table

Category	Exhibit
Antique	Zaveri
Modern	Johnsen
Classic	Terenida

Public Members

- Register() //A function to allow user to enter values

//Pno, Category and call FixExhibit() function

- ViewAll() //A function to display all the data members

Ans:

class Photo

```
{private:
    int Pno;
    char Category[20], Exhibit[20];
    void FixExhibit();
public:
    void Register();
    void ViewAll();
};

void Photo :: FixExhibit()
{if (strcmp(Category,"Antique")==0)
    Exhibit = "Zaveri";
else if (strcmp(Category,"Modern")==0)
    Exhibit = "Johnsen";
Else if (strcmp(Category,"Classic")==0)
    Exhibit = "Terenida";
}

void Photo :: Register()
{cin>>Pno;
    gets(Category);
    FixExhibit();
}

void Photo :: ViewAll()
```

```
{cout<<Pno<<Category<<Exhibit<<endl;  
}
```

(d) Answer the questions (i) to (iv) based on the following:

4 Marks

class Interior

```
{    int OrderId;  
    char Address[20];  
    protected:  
    float Advance;  
    public:  
    Interior();  
    void Book();  
    void View();  
};
```

class Painting:public Interior

```
{    int WallArea,ColorCode;  
    protected:  
    char Type;  
    public:  
    Painting();  
    void PBook();  
    void PView();  
};
```

```
class Billing : public Painting
```

```
{    float Charges;  
  
    void Calculate();  
  
    public: Billing();  
  
    void Bill();  
  
    void BillPrint();  
  
};
```

(i) Which type of Inheritance out of the following is illustrated in the above example ?

- Single Level Inheritance**
- Multi Level Inheritance**
- Multiple Inheritance**

Ans: In the above code fragment Multi Level Inheritance is used.

(ii) Write the names of all the data members, which are directly accessible from the member functions of class Painting.

Ans: WallArea, ColorCode, Type and Advance are the data members that are accessible from class Painting.

(iii) Write the names of all the member functions, which are directly accessible from an object of class Billing.

Ans: Bill(), BillPrint(), PBook(), PView(), Book() and View() are the data functions accessible from the object of class Billing. {Constructors are by default included in this list}

(iv) What will be the order of execution of the constructors, when an object of class Billing is declared ?

Ans: Interior() - Painting() - Billing() {Here the order matters}

3. (a) Write the definition of a function `Change(int P[], int N)` in C++, which should change all the multiples of 10 in the array to 10 and the rest of the elements as 1. 2 Marks

For example, if an array of 10 integers is as follows:

P[0]	P[1]	P[2]	P[3]	P[4]	P[5]	P[6]	P[7]	P[8]	P[9]
100	43	20	56	32	91	80	40	45	21

After executing the function, the array content should be changed as follows :

P[0]	P[1]	P[2]	P[3]	P[4]	P[5]	P[6]	P[7]	P[8]	P[9]
10	1	10	1	1	1	10	10	1	1

Ans:

```
void Alter (int P[], int size)
```

```
{for (int i=0 , i<size ,i++)
```

```
    if (P[i] %10 == 0)
```

```
        P[i] =10;
```

```
    else
```

```
        P[i] =1;
```

```
}
```

(b) A two dimensional array `ARR[50][20]` is stored in the memory along the row with each of its elements occupying 4 bytes. Find the address of the element `ARR[30][10]`, if the element `ARR[10][5]` is stored at the memory location 15000. 3 Marks

Ans:

Address of ARR[10][5]

$$15000 = \text{Base address} + W [(I - 1) * C + (J - 1)]$$

$$= \text{Base address} + 4(9 * 20 + 4)$$

$$= \text{Base address} + 4(184)$$

$$= \text{Base address} + 736$$

$$\text{Hence, Base address} = 15000 - 736$$

$$= 14264$$

Address of ARR[50][20]

$$= 14264 + 4[(30-1) * 20 + (10-1)]$$

$$= 14264 + 4[29 * 20 + 9]$$

$$= 14264 + 4[589]$$

$$= 16620$$

(c) Write the definition of a member function PUSH() in C++, to add a new book in a dynamic stack of BOOKS considering the following code is already included in the program: 4 Marks

struct BOOKS

{ char ISBN[20], TITLE[80];

BOOKS *Link;

};

class STACK

{ BOOKS *Top;

public:

STACK()

```
{Top=NULL;
}

void PUSH();
void POP();
~STACK();
};
```

Ans:

```
void STACK :: PUSH()
{BOOKS *temp;
temp = new BOOKS;
gets(temp ->ISBN);
gets(temp ->TITLE);
temp->Link = Top;
Top = temp;
}
```

(d) Write a function REVIEW(int P[][5],int N,int M) in C++ to display the content of a two dimensional array, with each row content in reverse order.

3 Marks

For example, if the content of array is as follows:

15	12	56	45	51
13	91	92	87	63
11	23	61	46	81

The function should display output as

51 45 56 12 15
63 87 92 91 13
81 46 61 23 81

Ans:

```
void REVROW(int P[][5] , int N, int M)
```

```
{ for (int r=0 ; r<N ;r++)
```

```
{     for(int c=M-1 ; c>=0 ; c--)
```

```
        cout<<P[r][c];
```

```
        cout<<endl;
```

```
    }
```

```
}
```

(e) Convert the following Infix expression to its equivalent Postfix expression, showing the stack contents for each step of conversion : 2 Marks

$U * V + R / (S - T)$

Ans:

$U * V + R / (S - T)$

$= ((U * V) + (R / (S - T)))$

Element	Stack	Postfix
(		
(		
U		U
*	*	

V		UV
)		UV*
+	+	
(		
R		UV*R
/	+/	
(		
S		UV*RS
-	+/-	
T		UV*RST
)		UV*RST-
)		UV*RST-/
)		UV*RST-/+

Therefore the required Postfix Expression is [UV*RST-/+]

4. (a) Write function definition for TOWER() in C++ to read the content of a text file WRITEUP.TXT, count the presence of word TOWER and display the number of occurrences of this word. 2 Marks

Note :

- The word **TOWER** should be an independent word
- Ignore type cases (i.e. lower/upper case)

Example : If the content of the file WRITEUP.TXT is as follows :

The Tower of Hanoi is an interesting problem. Mobile phone tower is away from here. Views from EIFFEL TOWER are amazing.

The function TOWER() should display the following :

3

Ans:

```
void TOWER()
{int COUNT=0;
  ifstream F (WRITEUP.TXT);
  char K[20];
  while (!F.eof())
  {    F>>K;
      If (strcmpi(K,"TOWER")==0)
          COUNT++;
  }
  cout<<COUNT;
  F.close();
}
```

(b) Write a definition for function COSTLY() in C++ to read each record of a binary file GIFTS.DAT, find and display those items, which are priced more than 2000. Assume that the file GIFTS.DAT is created with the help of objects of class GIFTS, which is defined below:

3 Marks

class GIFTS

```
{int CODE;

    char ITEM[20];

    float PRICE;

    public:

void Procure()
{    cin>>CODE;
        gets (ITEM);
        cin>>PRICE;
}

void View()
{    cout<<CODE<<": "<<ITEM<<": "<<PRICE<<endl;
}

float GetPrice()
    {return PRICE;
}.

};
```

Ans:

```
void COSTLY()
{GIFTS X;
    ifstream( "GIFTS.DAT",ios::binary);
    While (F.read((char*)&X,sizeof(X)))
{    if(X.GetPrice())>2000)
```

```

        X.View();

    }

    F.close();

}

```

(c) Find the output of the following C++ code considering that the binary file MEMBER.DAT exists on the hard disk with records of 100 members:

1 Mark

```

class MEMBER
{
    int Mno;

    char Name[20];

public:
    void In();
    void Out();
};

void main()
{
    fstream MF;
    MF.open("MEMBER.DAT",ios::binary|ios::in);
    MEMBER M;
    MF.read((char*)&M, sizeof(M));
    MF.read((char*)&M, sizeof(M));
    MF.read((char*)&M, sizeof(M));
    int POSITION= MF.tellg()/sizeof(M);
    cout<<"PRESENT RECORD:"<<POSITION<<endl;
}

```

```
MF.close();
```

```
}
```

Ans: The required output is:

PRESENT RECORD:3

SECTION - B

Only for candidates, who opted for Python

1. (a) How is `__init()` different from `__del()` ? **2 Marks**

Ans: `__init()` is the constructor of the class which is invoked automatically as the object of the class is created.

`__del()` is the destructor of the class which is invoked as soon as the scope of the object ends.

For example:

```
class Test :
```

```
def __init__(self):
```

```
self.data=10;
```

```
print('Data:',self.data,'created');
```

```
def __del__(self):
```

```
print('Data:',self.data,'deleted');
```

```
t = Test();
```

```
del t;
```

(b) Name the function/method required to **1 Mark**

(i) check if a string contains only alphabets

Ans: To do this isalpha() is used

(ii) give the total length of the list

Ans: To do this len() is used

**(c) Rewrite the following code in python after removing all syntax error(s).
Underline each correction done in the code. 2 Marks**

def Sum(Count) #Method to find sum

S=0

for I in Range(1,Count+1):

S+=I

RETURN S

print Sum[2] #Function Call

print Sum[5]

Ans:

def Sum(Count): #Method to find sum #1 Error

S=0

for I in range(1,Count+1): #1 Error

S+=I

return S #1 Error

print Sum(2) #Function Call #1 Error

print Sum(5) #1 Error

(d) Find and write the output of the following python code: 2 Marks

for Name in ['John', 'Garima','Seema','Karan']:

print Name

```

    if Name[0]=='S':
        break
    else:
        print 'Completed!'
print 'Well Done!'

```

Ans: The output of the above code fragment is:

John

Garima

Seema

Well Done!

(e) Find and write the output of the following python code:

2 Marks

```

class Emp:
    def __init__(self,code,nm): #constructor
        self.Code=code
        self.Name=nm
    def Manip(self):
        self.Code=self.Code+10
        self.Name='Karan'
    def Show(self,line):
        print self.Code,self.Name,line
s=Emp(25,'Mamta')
s.Show(1)

```

s.Manip()

s.Show(2)

print s.Code+len(s.Name)

Ans:

25 Mamta 1

35 Karan 2

40

(f) What are the possible outcome(s) executed from the following code ? Also specify the maximum and minimum values that can be assigned to the variable COUNT. 2 Marks

TEXT="CBSEONLINE"

COUNT=random.randint(0,3)

C=9

while TEXT[C]!='L':

print TEXT[C]+TEXT[COUNT]+'*',

COUNT=COUNT+1

C=C-1

(i)

(ii)

(iii)

(iv)

EC*NB*IS* NS*IE*LO*

ES*NE*IO*

LE*NO*ON*

Ans: (i) and (iii) are the possible outcomes of the above code fragment.

Min. and Max values of COUNT are 0 and 3

2. (a) Illustrate the concept of inheritance with the help of a python code.

2 Marks

Ans:

```
class Base:
    def __init__(self):
        print "Base Constructor"
    def show1(self):
        print "Base class"
class Der(Base):
    def __init__(self):
        Print "Derived Constructor"
    def show2(self):
        print "Derived class"
```

(b) What will be the output of the following python code ? Explain the try and except used in the code.

2 Marks

```
A=0
B=6
print 'One'
try:
    print 'Two'
    X=B/A
    Print 'Three'
except ZeroDivisionError:
```

```
print B*2
print 'Four'
except:
    print B*3
    print 'Five'
```

Ans: The output of the above code fragment is:

One
Two
12
Four

The code written within try invokes the exception written after except ZeroDivisionError: in a situation there is a division by zero error otherwise the default exception is executed.

(c) Write a class PHOTO in Python with following specifications: 4 Marks

Instance Attributes

- Pno # Numeric value
- Category # String Value
- Exhibit # Exhibition Gallery with String value

Methods:

- FixExhibit() #A method to assign #Exhibition Gallery as per Category

#as shown in the following table

Category	Exhibit
Antique	Zaveri

Modern	Johnsen
Classic	Terenida

Category Exhibit Antique Zaveri Modern Johnsen Classic Terenida

– Register() **#A function to allow user to enter values**

#Pno, Category and call FixExhibit() method

– ViewAll() **#A function to display all the data members**

Ans:

class PHOTO:

Pno=0

Category=""

Exhibit=""

def FixExhibit():

If self.Category=="Antique":

self.Exhibit="Zaveri"

elif self.Category=="Modern":

self.Category=="Johnsen"

Elif self.Category=="Classic":

self.Exhibit="Terenida"

def Register():

self.Pno=int(input("Enter Pno:"))

self.Category=input("Enter Name:")

```
self.FixExhibit()
```

```
Def ViewAll()
```

```
print self.Pno,self.Category,self.Exhibit
```

(d) What is operator overloading with methods ? Illustrate with the help of an example using a python code.. **2 Marks**

Ans: Operator overloading as the name suggests is the method to give extended meaning to the predefined meaning of the operator. For instance, '+' operator is used to add two numbers and also to concatenate two strings.

For eg:

```
x=5
```

```
y=9
```

```
print (x+y)          #'+' here adds two numbers
```

```
x='Ram'
```

```
y='Kumar'
```

```
print (x+y)          #'+' here concatenates two strings
```

(e) Write a method in python to display the elements of list twice, if it is a number and display the element terminated with '*' if it is not a number.

2 Marks

For example, if the content of list is as follows :

```
MyList=['RAMAN','21','YOGRAJ','3','TARA']
```

The output should be

```
RAMAN*
```

```
2121
```

YOGRAJ*

33

TARA*

Ans: def meth(U):

for x in U:

 If x.isnumeric():

 print(2*x)

 else:

 print(x+'*')

3. (a) What will be the status of the following list after the fourth pass of bubble sort and fourth pass of selection sort used for arranging the following elements in descending order ? **3 Marks**

34,-6,12,-3,45,25

Ans: After the fourth pass

In bubble sort:

45,34,25,12,-3,-6

In selection sort:

45,34,25,12,-6,-3

(b) Write a method in python to search for a value in a given list (assuming that the elements in list are in ascending order) with the help of Binary Search method. The method should return -1, if the value is not present else it should return the position of the value present in the list. **2 Marks**

Ans:

```
def BS( U , K):
    L=0
    H=len(U)-1
    flag=false
    while (L <= H) and (not flag):
        M = (L+H)/2
        if L[M] == K:
            flag=true
        elif L[M] < K:
            L = M+1
        else:
            H = M-1
    if flag:
        return M+1
    else:
        return -1
```

(c) Write PUSH(Names) and POP(Names) methods in python to add Names and Remove names considering them to act as Push and Pop operations of Stack.

2 Marks

Ans:

```
def PUSH(Names):
    Stack.append(Names)
    print 'Element:',Name,'Pushed'
```

Def POP(Names):

```
if Stack == []:
    print('Stack is empty.....')
else:
    print('Popped element is', Stack.pop())
```

(d) Write a method in python to find and display the composite numbers between 2 to N. Pass N as an argument to the method. 2 Marks

Ans:

```
def com_num(N):
    for X in range(2,N+1):
        MID = X/2
        for Y in range(2, MID+1):
            If X%Y ==0:
                print(X)
                break
```

(e) Evaluate the following postfix notation of expression. Show status of stack after every operation. 2 Marks

34,23,+,4,5,*,-

Ans:

Element	Stack
34	34
23	34,23
+	57

4	57,4
5	57,4,5
*	57,20
-	37

4. (a) Differentiate between the following :

1 Mark

(i) `f = open('diary.txt', 'a')`

(ii) `f = open('diary.txt', 'w')`

Ans: In (i) dairy.txt is opened in append mode ie at the end of the file, on the other hand in (ii)

dairy .txt is opened in writing mode and hence data here is written in the file right from the beginning.

(b) Write a method in python to read the content from a text file

story.txt line by line and display the same on screen.

2 Marks

Ans:

```
def READ():
```

```
    F = open('story.txt', 'r')
```

```
    For line in F:
```

```
        print line
```

(c) Consider the following definition of class Student. Write a method

in python to write the content in a pickled file student.dat.

2 Marks

class Student:

```
def __init__(self,A,N):
```

```
self.Admno=A
```

```
self.Name=N
```

```
def Show(self):
```

```
print(self.Admno,"#",self.Name)
```

Ans:

```
def write_data(self):
```

```
PF = open('student.dat','wb')
```

```
pickle.dump(self,PF)
```

```
PF.close()
```

SECTION C

For all candidates

5. (a) Observe the following table carefully and write the names of the most appropriate columns, which can be considered as

(i) candidate keys and (ii) primary key :

2 Marks

Code	Item	Qty	Price	Transaction Date
1001	Plastic Folder 14"	100	3400	2014-12-14
1004	Pen Stand Standard	200	4500	2015-01-31
1005	Stapler Mini	250	1200	2015-02-28
1009	Punching Machine Small	200	1400	2015-03-12
1003	Stapler Big	100	1500	2015-02-02

Ans:

(i) Candidate Keys are Code & Item

(ii) Primary Key is Code

(b) Consider the following DEPT and EMPLOYEE tables. Write SQL queries for (i) to (iv) and find outputs for SQL queries (v) to (viii). 6 Marks

Table: DEPT

DCODE	DEPARTMENT	LOCATION
D01	INFRASTRUCTURE	DELHI
D02	MARKETING	DELHI
D03	MEDIA	MUMBAI
D05	FINANCE	KOLKATA
D04	HUMAN RESOURCE	MUMBAI

Table: EMPLOYEE

ENO	NAME	DOJ	DOB	GENDER	DCODE
1001	George K	2013-09-02	1991-09-01	MALE	D01
1002	Ryma Sen	2012-12-11	1990-12-15	FEMALE	D03
1003	Mohitesh	2013-02-03	1987-09-04	MALE	D05
1007	Anil Jha	2014-01-17	1984-10-19	MALE	D04
1004	Manila Sahai	2012-12-09	1986-11-14	FEMALE	D01
1005	R Sahay	2013-11-18	1987-03-31	MALE	D02
1006	Jaya Priya	2014-06-09	1985-06-23	Female	D05

Note : DOJ refers to date of joining and DOB refers to date of Birth of employees.

(i) To display Eno, Name, Gender from the table EMPLOYEE in ascending order of Eno.

Ans: SELECT Eno,Name,Gender FROM Employee ORDER BY Eno;

(ii) To display the Name of all the MALE employees from the table EMPLOYEE.

Ans: SELECT Name FROM Employee WHERE Gender = 'MALE';

(iii) To display the Eno and Name of those employees from the table EMPLOYEE who are born between '1987-01-01' and '1991-12-01'.

Ans: SELECT Eno, Name FROM Employee WHERE DOB BETWEEN '1987-01-01' AND '1991-12-01';

(iv) To count and display FEMALE employees who have joined after '1986-01-01'.

Ans: SELECT COUNT(*) FROM Employee WHERE Gender='FEMALE' AND DOJ > '1986-01-01';

(v) SELECT COUNT(*),DCODE FROM EMPLOYEE GROUP BY DCODE HAVING COUNT(*)>1;

Ans:

COUNT	DCODE
2	D01
2	D05

(vi) SELECT DISTINCT DEPARTMENT FROM DEPT;

Ans: DEPARTMENT

INFRASTRUCTURE

MARKETING

MEDIA

FINANCE

HUMAN RESOURCE

**(vii) SELECT NAME,DEPARTMENT FROM EMPLOYEE E,DEPT D
WHERE E.DCODE=D.DCODE AND ENO<1003;**

Ans: NAME DEPARTMENT

George K INFRASTRUCTURE

Ryma Sen MEDIA

(viii) SELECT MAX(DOJ), MIN(DOB) FROM EMPLOYEE;

Ans: MAX(DOJ) MIN(DOB)

2014-06-09 1984-10-19

6. (a) Verify the following using Boolean Laws :

2 Marks

$$U' + V = U'V' + U'.V + U.V$$

Ans:

Taking L.H.S:

$$U' + V$$

$$= U'.(V + V') + V.(U' + U)$$

$$= U'.V + U'.V' + U'.V + U.V$$

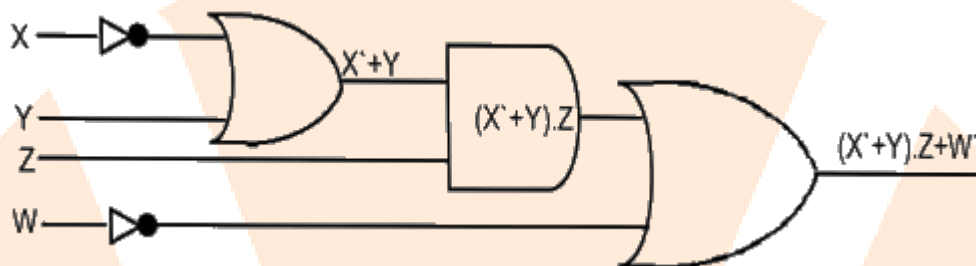
$$= U'.V + U'.V' + U.V$$

Hence , L.H.S=R.H.S

(b) Draw the Logic Circuit for the following Boolean Expression: 2 Marks

$$(X' + Y).Z + W'$$

Ans:



(c) Derive a Canonical POS expression for a Boolean function F, represented by the following truth table :

P	Q	R	F(P,Q,R)
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Ans: $F(P, Q, R) = (P + Q + R')(P + Q' + R)(P' + Q + R')(P' + Q' + R)$

(d) Reduce the following Boolean Expression to its simplest form using K-Map: 2 Marks

$$F(X, Y, Z, W) = (0, 1, 4, 5, 6, 7, 8, 9, 11, 15)$$

Ans: The K-Map of the given boolean expression is given below:

	$X'Y'$	$X'Y$	XY	XY'
$Z'W'$	1	1		1
$Z'W$	1	1		1
ZW		1	1	1
ZW'		1		

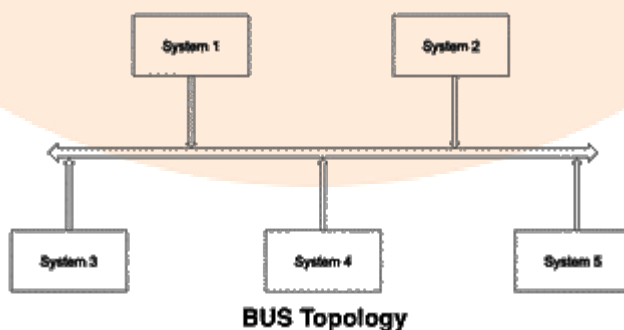
Or

	$Z'W'$	$Z'W$	ZW	ZW'
$X'Y'$	1	1		
$X'Y$	1	1	1	1
XY			1	
XY'	1	1	1	

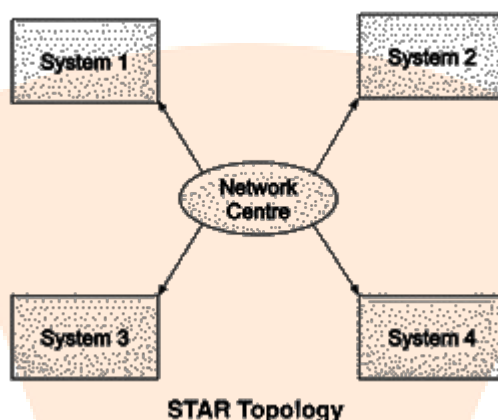
Hence, simplified boolean expression is $Y'Z' + X'Y + XZW$

7. (a) Illustrate the layout for connecting computers in a Bus and a Star topology of Networks. 1 Mark

Ans: The layout for Bus Topology is as given below:



The layout for Star Topology network is as given below:



(b) What kind of data gets stored in cookies and how is it useful ? 1 Mark

Ans: By the use of the cookies, the server of a website receives the information regarding the browser which is then stored in the hard drive in the form of a text file. Through this the server remembers the information about the visited sites.

(c) Differentiate between packet switching over message switching ? 1 Mark

Ans:

Packet Switching	Message Switching
It follows the store and forward principle for fixed packets.	It follows the store and forward principle for complete message.
It fixes the upper limit for packet size.	No limit on block size.

(d) Out of the following, which is the fastest (i) wired and

(ii) wireless medium of communication ?

1 Mark

Infrared, Coaxial Cable, Ethernet Cable, Microwave, Optical Fiber

Ans:

(i) Wired : Optical Fiber

(ii) Wireless : Infrared / Microwave

(e) What is a Trojan Horse ?

1 Mark

Ans:

- A Trojan horse is a type of malware that is often disguised as legitimate software.
- It can be employed by hackers aiming to gain access to users' systems.
- Users are usually tricked by some form of social engineering into loading and executing
- Trojans on their systems. Once activated, they can enable cyber-criminals to spy,
- steal sensitive data, and gain backdoor access to the system.

(f) Out of the following, which all comes under cyber crime ?

1 Mark

(i) Stealing away a brand new hard disk from a showroom.

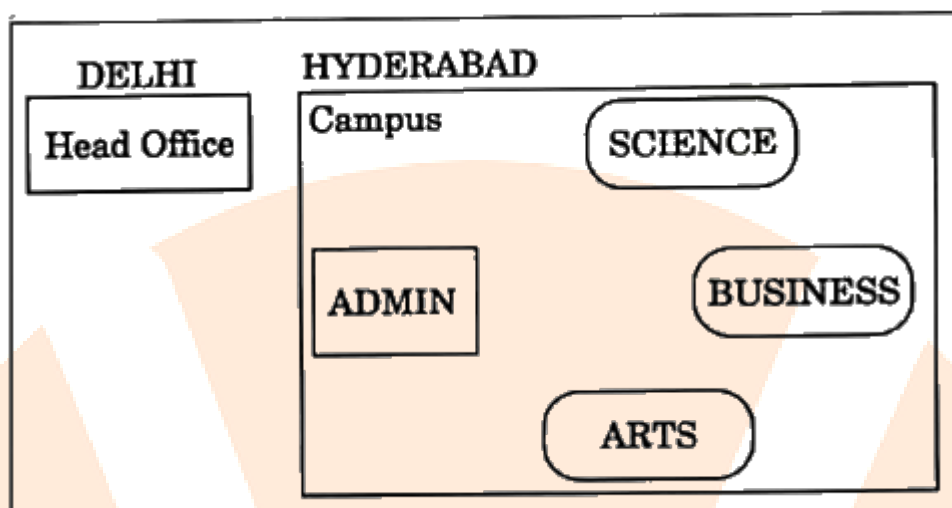
(ii) Getting in someone's social networking account without his consent and posting on his behalf.

(iii) Secretly copying data from the server of an organization and selling it to the other organization.

(iv) Looking at online activities of a friend's blog.

Ans: Of the given options, (ii) & (iii) come under the cyber crime

(g) Xcelencia Edu Services Ltd. is an educational organization. It is planning to set up its India campus at Hyderabad with its head office at Delhi. The Hyderabad campus has 4 main buildings - ADMIN, SCIENCE, BUSINESS and ARTS. You as a network expert have to suggest the best network related solutions for their problems raised in (i) to (iv), keeping in mind the distances between the buildings and other given parameters.



Shortest Distance between various buildings:

ADMIN to SCIENCE	65M
ADMIN to BUSINESS	100M
ADMIN to ARTS	60M
SCIENCE to BUSINESS	75M
SCIENCE to ARTS	60M
BUSINESS to ARTS	50M
DELHI HEAD OFFICE to HYDERABAD CAMPUS	1600KM

Number Of Computers installed at various building are as follows:

ADMIN	100
SCIENCE	85
BUSINESS	40
ARTS	12

**DELHI HEAD
OFFICE**

20

(i) Suggest the most appropriate location of the server inside the HYDERABAD campus (out of

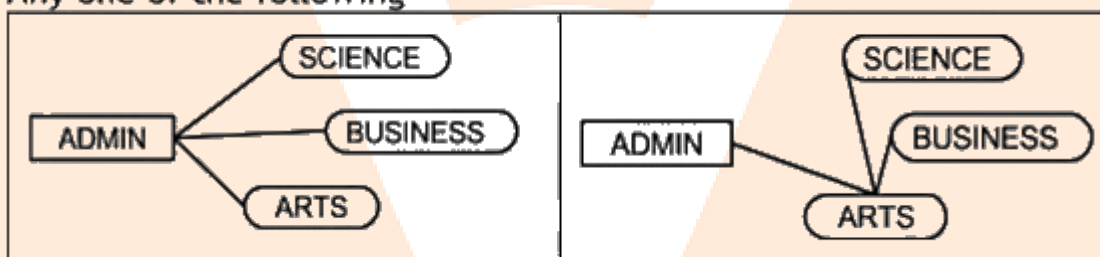
the 4 buildings), to get the best connectivity for maximum number of computers. Justify your answer. 1 Mark

Ans: Location of server could be either in ADMIN since it has maximum number of computers or it could be in ARTS since it is close to all other buildings.

(ii) Suggest and draw the cable layout to efficiently connect various buildings within the HYDERABAD campus for connecting the computers. 1 Mark

Ans:

Any one of the following



(iii) Which hardware device will you suggest to be procured by the company to be installed to protect and control the internet uses within the campus ? 1 Mark

Ans: Company could use either Firewall or Router.

(iv) Which of the following will you suggest to establish the online face-to-face communication between the people in the Admin Office of HYDERABAD campus and DELHI Head Office? 1 Mark

(i) E-mail

(ii) Text Chat

(iii) Video Conferencing

(iv) Cable TV

Ans: For face-to-face communication it can use video-conferencing.

